

Answer Key — The Three States of Matter

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1 **(b) liquid** — liquids keep their volume but take the container's shape.
-
- Q2 **(c) solids** — solids have the strongest interparticle attractions.
-
- Q3 **(b) melting point** — that temperature is the melting point.
-
- Q4 **(b) fluids** — both flow, so they are fluids.
-
- Q5 **(a) Both true, R explains A** — negligible forces let gas particles spread out fully — exactly as R says.
-
- Q6 **evaporation** — surface vaporisation below boiling is evaporation.
-
- Q7 **vibrate** — solid particles vibrate/oscillate in place.
-

Short answer

- Q8 Solids: definite shape and volume. Liquids: definite volume, no fixed shape. Gases: no fixed shape or volume.
-
- Q9 Melting point: temperature at which a solid melts to liquid at atmospheric pressure. Boiling point: temperature at which a liquid boils to vapour at atmospheric pressure.
-
- Q10 Boiling occurs at the boiling point, throughout the liquid (with bubbles); evaporation is slow, only at the surface, and happens at any temperature below boiling.
-

Long / application

- Q11 Heating makes the particles vibrate more. When vibrations are vigorous enough the particles leave their fixed positions and the interparticle forces weaken — the solid melts to a liquid (at its melting point). On further heating, particles move so fast they escape the liquid; forces become negligible and it turns to gas (at the boiling point).
-
- Q12 Liquid particles move freely, so the water takes each container's shape (no fixed shape), but the particles stay close together so the amount stays the same (definite volume of 200 mL). Hence liquids have a definite volume but no fixed shape.
-